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The Acquisition and Extinction of Fear of Painful Touch: A Novel Tactile Fear Conditioning Paradigm

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Abstract: **Fear of touch, due to allodynia and spontaneous pain, is not well understood. Experimental methods to advance this topic are lacking, and therefore we propose a novel tactile conditioning paradigm. Seventy-six pain-free participants underwent acquisition in a predictable as well as an unpredictable pain context. In the predictable context, vibrotactile stimulation was paired with painful electrocutaneous stimulation (simulating allodynia). In the unpredictable context, vibrotactile stimulation was unpaired with pain (simulating spontaneous pain). During an extinction phase, a cue exposure and context exposure group continued in the predictable and unpredictable context, respectively, without pain. A control group received continued acquisition in both contexts. Self-reported fear and skin conductance responses, but not startle responses, showed fear of touch was acquired in the predictable context. Context-related startle responses showed contextual fear emerged in the unpredictable context, together with elevated self-reported fear and skin conductance responses evoked by the unpaired vibrotactile stimulations. Cue exposure reduced fear of touch, whereas context exposure reduced contextual fear. Thus, painful touch leads to increased fear, as does touch in the same context as unpredictable pain, and extinction protocols can reduce this fear. We conclude that tactile conditioning is valuable for investigating fear of touch and can advance our understanding of chronic pain.**

Perspectives: ***The acquisition and extinction of fear of touch was investigated in a clinical analog study using a novel tactile fear conditioning paradigm. The results have implications for research on the development and treatment of chronic pain conditions characterized by allodynia and spontaneous pain fluctuations.***

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Key words: *Pain-related fear, fear conditioning, extinction, touch, vibrotactile.*

Chronic pain patients are a highly diverse group, experiencing a diverse range of pain symptoms. Research on the mechanisms of these pain experiences has generally divided them into categories of stimulus-evoked pain and stimulus-independent pain,[3,42] a distinction also reflected in the assessment of neuropathic pain syndromes.[1] Stimulus-evoked pain includes symptoms such as allodynia (perceiving nonpainful touch as painful), whereas stimulus-independent pain can be an ongoing or fluctuating sensation, such as burning, tingling, or stabbing, etc.[5] The stimulus-evoked pain may

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be experienced by chronic pain patients as predictable; in the case of allodynia it will be expected that touch to the affected area will be painful. In the case of stimulus-independent pain there are no clear cues or indicators of pain fluctuations, from the patient's perspective the pain occurs unpredictably.

Interestingly, a similar dichotomy exists within research using fear conditioning paradigms to investigate anxiety disorders.[9] Manipulations of predictability, for painful as well as nonpainful aversive events, have been used in classical conditioning paradigms to elicit cued and contextual fear[36]: cued fear is a phasic response to an explicit cue that has been paired with pain, and is predictive of its occurrence. However, when there is no cue predicting the occurrence of pain (it is unpredictable) then the context becomes predictive of threat and a chronic anticipatory anxious state is evoked.[9]

Fear is a core aspect of chronic pain; the (pain-related) fear experienced by chronic pain patients motivates maladaptive protective behaviors, such as avoiding touch to the affected limb, which maintains a vicious cycle of pain, avoidance, and disability.[40,41] Therefore, investigating how different types of pain may lead to different types of fear, and thus motivate different protective behaviors, would have implications for our understanding of how chronic pain disability develops, is maintained, and could be treated.

To study these processes in the laboratory, we introduce a novel tactile fear conditioning paradigm that models the experience of stimulus-evoked as well as stimulus-independent pain. Allodynia was modeled in a predictable pain context by pairing nonpainful touch (vibrotactile stimulation) with painful electrocutaneous stimulation. In an unpredictable pain context, stimulus-independent pain is modeled using unpaired presentations of nonpainful touch and pain. An extinction protocol is also included, during which the nonpainful touch and (un)predictable contexts were presented, but the pain was omitted. This models exposure to feared touch, and feared contexts, that occurs in exposure-based treatments for chronic pain.[23]

We hypothesized that: 1) in the predictable context, fear responses would be higher for the vibrotactile stimulation that was consistently followed by pain than for vibrotactile stimulation that was never paired with pain, 2) more contextual fear would occur in the unpredictable context than in the predictable context, and 3) fear responses would be reduced when pain was omitted in both contexts. Three responses served as a proxy for fear learning: self-reported fear, skin conductance responses (SCRs), and eyeblink startle potentiation. A Bayesian approach to hypothesis testing was used. This has several advantages over a frequentist approach,[38] including moving beyond rejection of a null hypothesis and instead quantifying the degree of support for multiple competing hypotheses.

Methods

Participants

Eighty-two participants were recruited and provided informed consent; they were compensated with 2 course credits or €16. The Ethical Committee of the University of Leuven approved the experimental protocol (S56496). Exclusion criteria were pregnancy, cardiovascular, neurological, or pulmonary disease, electronic implants, current/past clinical depression, current/past anxiety, current acute pain, current/past chronic pain, hearing loss, or color blindness. Six data sets were incomplete because of equipment failure (n = 4) and dropout (n = 2). The statistical analyses were run on the remaining data (N = 76; mean = 22.6 years, range = 18–59 years; 54 female; 64 right-handed). This sample size has been shown to be adequate for a similar paradigm.[27]

Experimental Stimuli

To simulate innocuous touch a modified refreshable braille device (Metec AG, Stuttgart, Germany) generated vibrotactile stimulation. This device consists of 4 $6 \times 16 \times 74$-mm modules, each with a 2×4 array of plastic pins (2.45-mm spacing) that raise and lower in a random pattern at a frequency of 30 Hz. The modules were placed under the fingertips of the index (D2) and little fingers (D5) of the right and left hand. Fingers were always stimulated separately (4-second duration). The stimulation of each finger served as a conditioned stimulus (CS) in the conditioning procedure, of which there were 4 types. The finger and CS type assignment was counterbalanced.

Also in the conditioning procedure were 2 painful unconditioned stimuli (pain-USs; predictable and unpredictable), which were delivered to opposite wrists (wrist/pain-US type was counterbalanced). These pain-USs were electrocutaneous stimulation (2-ms duration) delivered through a constant current stimulator each (DS7A; Digitimer, Welwyn Garden City, England), and 8-mm Ag/AgCl (silver-chloride) electrodes filled with K-Y gel (Johnson & Johnson, New Brunswick, NJ) attached to the left and right wrist. Per individual, we set the subjective intensity level for both pain-USs as "painful and demanding some effort to tolerate" using an ascending stepwise calibration procedure for the right and then the left wrist, carried out before the start of the conditioning experiment. This procedure required participants to rate each pain-US on a scale from 0 ("no sensation at all") to 10 ("worst pain imaginable"), with an 8 corresponding to "painful and demanding some effort to tolerate." Note that the average physical intensity of the pain-US was comparable for the left (mean = 29.5 ± 16.9 mA) and right (mean = 29.2 ± 18.5 mA), as was subjective intensity throughout the experiment (predictable: mean = 7.14 ± 1.60; unpredictable: mean = 7.22 ± 1.63).

The background color (orange/blue) of the computer screen indicated the experimental context (predictable/unpredictable pain), which was continuously present throughout a block of trials (Fig 1). The color and context pairing was counterbalanced.

Procedure

Preconditioning Phase

For familiarization with the CSs, habituation to the auditory startle probes, and recording of baseline

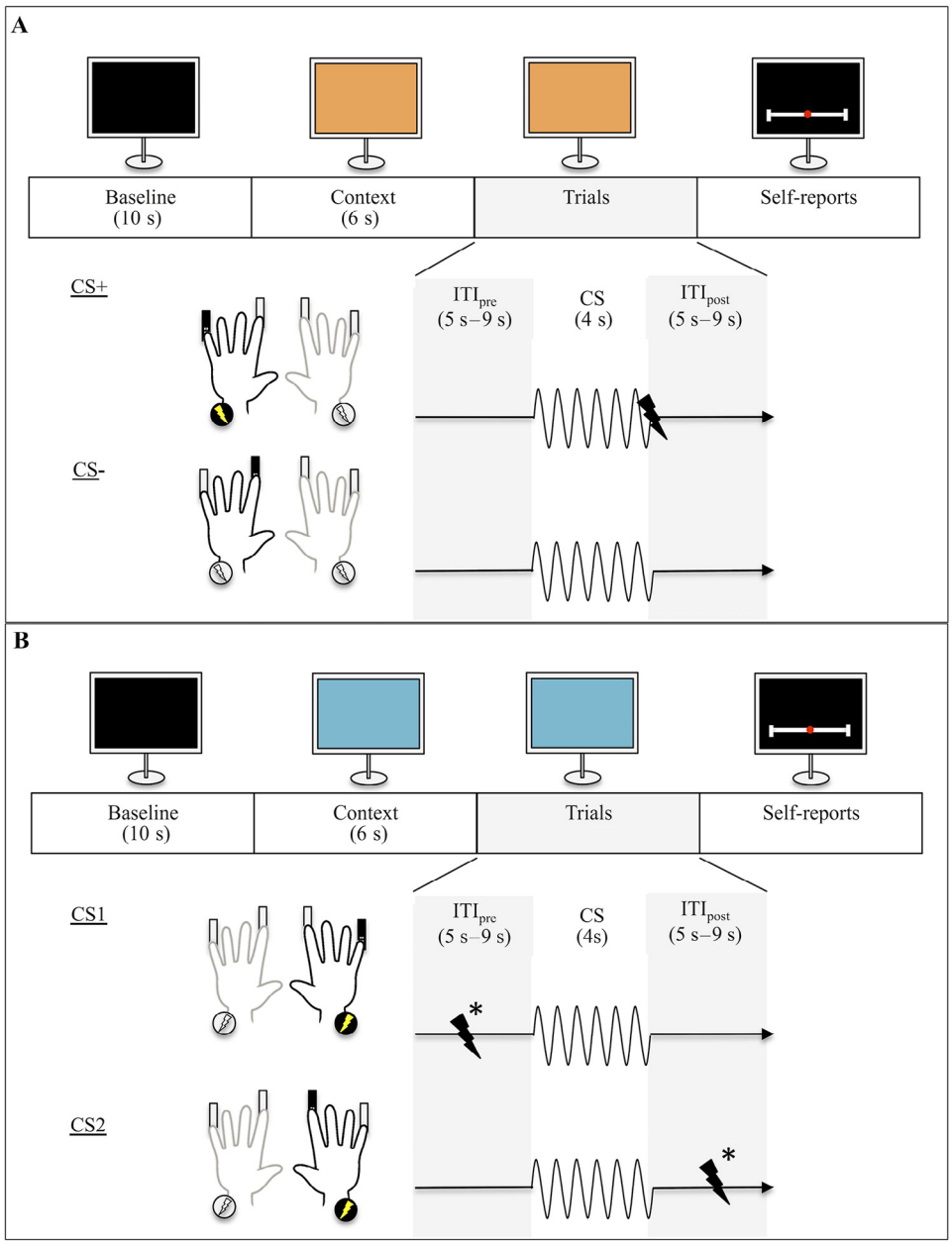

Figure 1. Overview of an exemplary trial structure for the predictable context (**A**) and unpredictable context (**B**). During the experiment participants sat in front of a computer screen (color names are for illustrative purposes to indicate screen color, but were not shown to the participant) with their hands in the position depicted in panel **A** and **B**. The black stimulation module indicates the finger being stimulated for each CS type. The lightning bolt in the black circle indicates the location of the pain-US, and its possible timing in the trial timeline. Note in the unpredictable context the pain-US could occur at a randomly selected second in the ITI, excluding 0 seconds and the last second of the ITI. Each block contained 10 trials (5 of each type). *Possible pain-US occurrence.

psychophysiological responses to the CSs and contexts, participants were presented with each of the CSs 5 times (Table 1). Note that no pain-USs were presented in this phase. The CS stimulations were presented in an intermixed order (ie, no more than 2 consecutive stimulations of the same finger) in 2 blocks, each with a different colored screen and stimulation restricted to fingers of either the left or the right hand. Auditory startle probes were delivered on each trial (to elicit the eyeblink startle response, see the Eyeblink Startle Response section in the section on Outcome Measures), 2 during each of the CSs and 3 in the intertrial interval (ITI) surrounding each of the CSs (20 probes in total). The ITI length varied between 5 and 9 seconds and occurred before (ITI_{pre}) as well as after the CS presentation (ITI_{post}).

Acquisition Phase

Participants were instructed to attend to the different types of stimuli presented, and that they would be asked about the relationships between stimuli. Eight blocks of trials were presented (4 predictable and 4 unpredictable). The order was semirandomized with no more than 2 consecutive blocks of the same type. To clearly separate blocks, and minimize carryover effects from one

Table 1. Study Design

	Preconditioning Within Subjects; 5 Trials Per CS Type	*Acquisition Within Subjects; 20 Trials Per CS Type*	*Extinction Between Subjects 20 Trials Per CS Type*	
Control group	(1 × Predictable context) 5 × CS+ 5 × CS–	(4 × Predictable context) 5 × CS+ 5 × CS–	(4 × Predictable context) 5 × CS+ 5 × CS–	(4 × Unpredictable context) 5 × CS1 5 × CS2
Cue exposure group	(1 × Unpredictable context) 5 × CS1 5 × CS2	(4 × Unpredictable context) 5 × CS1 5 × CS2	(4 × Predictable context) 5 × CS+ 5 × CS–	
Context exposure group				(4 × Unpredictable context) 5 × CS1 5 × CS2

context to another, each block began with a baseline period of 12 seconds during which only a black background screen was shown, followed by 6 seconds of only the context (computer background color). Subsequently, 10 trials were presented while the context remained visible. Each trial began with an ITI_{pre} (variable length: 5–9 seconds), followed by a CS, and then an ITI_{post} (variable length: 5–9 seconds). At the end of each block, the context was removed and rating scales were presented to assess fear of touch (see the Self-Reports section of the section on Outcome Measures). In the predictable blocks, 2 CS paired with a predictable pain-US (CS+) (eg, vibrotactile stimulation of left D5) and 5 CS never paired with a pain-US (CS–) (eg, stimulation of D2 on the same hand) were presented. The CS+ coterminated with the predictable pain-US (eg, vibrotactile stimulation of left D5) in 80% of trials, whereas the CS– was never paired with a pain-US. During the unpredictable context, stimulation occurred on the hand opposite to the stimulation in the predictable context. Five CS presented in an unpredictable pain context (CS1; eg, vibrotactile stimulation of right D5) and 2 CS presented in an unpredictable pain context (CS2; eg, vibrotactile stimulation of right D2) were presented. The unpredictable pain-US (ipsilateral to CS1/CS2) occurred during the ITI, not related to either CS1 or CS2, with an equal number of occurrences as in the predictable context. The unpredictable pain-USs were distributed so that one occurred at a random point in the ITI_{pre} and ITI_{post} of CS1 as well as CS2. The startle probe was delivered using the same timing as in the preconditioning phase.

Extinction Phase

We used a between-subjects extinction manipulation including 3 groups: the cue exposure group continued with 4 blocks in the predictable context without reinforcement of the CS+ (ie, no predictable pain-US presentation); the context exposure group continued with 4 blocks in the unpredictable context, without any unpredictable pain-US presentations; and the control group received continued acquisition training (4 blocks in the unpredictable context and 4 blocks in the predictable context), with presentation of the predictable as well as unpredictable pain-US. There was no change in instructions or explicit change in phase from the participant's perspective, irrespective of group. Group allocation was randomly assigned at the start of the experiment.

Outcome Measures

Self-Reports

To assess self-reported fear of the CSs, at the end of each block the following question was presented on the computer screen: "How afraid did you feel during the stimulation of your [finger] during the previous block?" Participants could respond using a foot pedal (USB triple foot switch, Delcom Products, Danbury, CT) that moved a cursor along a horizontal 11-point visual analog scale. This method was used to avoid movement of the hands from the stimulation modules, as well as possible disruption of the electrodes used for physiological measures and administration of the pain-US. The intensity of the CSs was also assessed ("How intense did you find the stimulation of your [finger] during the previous block?"). Intensity ratings were acquired to assess participants' experience of this new device, and are summarized in Supplementary Appendix 1. A question was also presented about the pain-USs ("How [painful] did you find the electrical stimulation during the previous block?"), as a measure of subjective pain-US experience. At the end of each phase (preconditioning, acquisition, and extinction) pain-US expectancy was assessed, as a retrospective contingency awareness check ("To what degree did you expect electrical stimulation with the stimulation of your [finger]?").

SCR

Autonomic arousal is a common part of the conditioned fear response and was therefore measured as the SCR.[24] SCR was measured throughout the experiment by passing 0.5 V between 2 electrodes (8 mm Ag/AgCl, filled with [K-Y] gel [Johnson & Johnson, New Brunswick, NJ]), placed on the hypothenar eminence of the left hand, using the Coulbourn Lab Linc V system (Coulbourn Instruments, Whitehall, PA) and a National Instruments data acquisition card (National Instruments Corp, Austin, TX).

Eyeblink Startle Response

The potentiation of the eyeblink startle response is a robust measure of conditioned fear[16,24] and was used in

this study as an index of fear of touch, when elicited during the CS presentation (either 1, 2, or 3 seconds after CS onset). When the startle response is elicited during the ITI it is used as a measure of contextual fear.[18] Electromyographic activity elicited by auditory probes (100 dB burst of white noise with instantaneous rise time, presented for 50 ms binaurally through headphones [Sennheiser HD 280 pro; Sennheiser Electronic GMBH & Co. KG, Wedemark, Germany]) was measured using 2 electrodes (4 mm Ag/AgCl electrodes, filled with electrolyte gel) placed under the left eye and 1 (ground electrode) on the forehead, after scrubbing of the skin to reduce impedance.[4] The signal was recorded (sampling rate = 1 kHz) between 500 ms before to 1,000 ms after probe onset and was low pass filtered (500 Hz), using the Coulbourn Lab Linc V system (Coulbourn Instruments) and a National Instruments data acquisition card.

Data Reduction and Analysis

Physiological Signal Preprocessing

The startle response data were visually inspected and trials were rejected from further analysis if visual inspection revealed artifacts (eg, elevated baseline value after spontaneous blink), in total 7.18% of trials. Nonresponse trials made up 8.30% of the data set. No significant differences in results were found when these trials were included or excluded, therefore the reported results include these data. A criterion of >50% nonresponse trials was used to classify 'nonresponders' and no participants reached this threshold. Peaks were defined as the maximum value within 21 and 175 ms after probe onset, whereas the baseline value was defined as the average of the first 20 ms after probe onset (using Psychophysiological Analysis software version 4.0[11]). The startle response was calculated by subtracting the baseline value from the peak and then z-transformed per participant (excluding the preconditioning trials).

The skin conductance data were visually inspected for artifacts (replaced using cubic spline interpolation), downsampled to 10 Hz, and smoothed. Using a continuous decomposition analysis the signal was decomposed into a tonic and phasic estimation of activity and the size of the phasic, event-related, activations were calculated during the response window (1–4 seconds after CS onset; for a full explanation of the method see Benedek and Kaernbach[2]). This event-related activity was z-transformed per participant (across all phases of the experiment).

Bayesian Hypothesis Testing Approach

Whereas traditional frequentist statistics assess the probability of observing a certain result under the null hypothesis (and use the *P* value to decide whether or not to accept this null hypothesis). Bayesian approaches can quantify the support for not only the null hypothesis but also multiple alternative hypotheses. The benefit is that the statistic more accurately reflects the answer to the experimental question: how likely is it that the proposed hypothesis is true, given the data that were collected? We used a Bayesian statistical approach implemented in the free software BIEMS (version 2012) (http://informative-hypotheses.sites.uu.nl/software/biems/).[32,33] In this approach, each hypothesis is effectively a model, where equality (=) and inequality (<>) constraints limit where the mean of a particular variable could occur relative to the means of the other variables included in the data set. In addition, a null hypothesis M0 (model 0) is always included, this unconstrained model simply states that the means exist but does not specify any relationship between them. The likelihood of each model given the data is calculated, and is combined with an index of the model complexity and fit to create a Bayes factor (BF) per model (BF <0 indicates the M0 is most likely), these BFs can then be compared to quantify which model is more likely to be true. This comparison is evident in the posterior model probabilities (PMP); these are the relative probabilities of each model included in the analysis (eg, the sum of all the BFs divided by the number of models, including M0). Although the hypotheses tested are different, a frequentist analysis is also included in Supplementary Appendix 2 to allow for an informal comparison of the 2 approaches.

Results

For each phase the hypotheses tested are described in the corresponding section, whereas the PMPs are shown in Figs 2 through 5. The PMPs for the hypotheses receiving the most support are reported in the main text whereas the PMPs for the competing hypotheses, as well as additional statistics (BFs) are in Supplementary Tables 1 through 6.

Preconditioning Phase

There were no competing hypotheses for the preconditioning phase and so all models were compared only with the null hypothesis (M0, there is no relationship between CSs). Participants reported no difference in self-reported pain-US expectancy (BF = 4,572.75, PMP = 1.0, compared with M0) or fear of touch (BF = 2,059.72, PMP = 1.0, compared with M0), SCR (BF = 546.76, PMP = 1.0, compared with M0), or startle response (BF = 31.54, PMP = .97, compared with M0) in response to the different CSs. There was also no difference in ITI startle response during the predictable and unpredictable context (BF = 3.29, PMP = .77, compared with M0). However, the startle response evoked during the CSs (mean = –.27, SD = .05) was considerably lower than that evoked during the ITI (mean = .19, SD = .02; BF = 117.41, PMP = .99, compared with M0). See supplementary materials for further discussion.

Acquisition Phase

To compare fear responses (self-reports, SCR, and startle response) to the CSs, the data from this phase were used to compare 3 alternative hypotheses and 2 null hypotheses (no relationship between CSs [M0] and equal

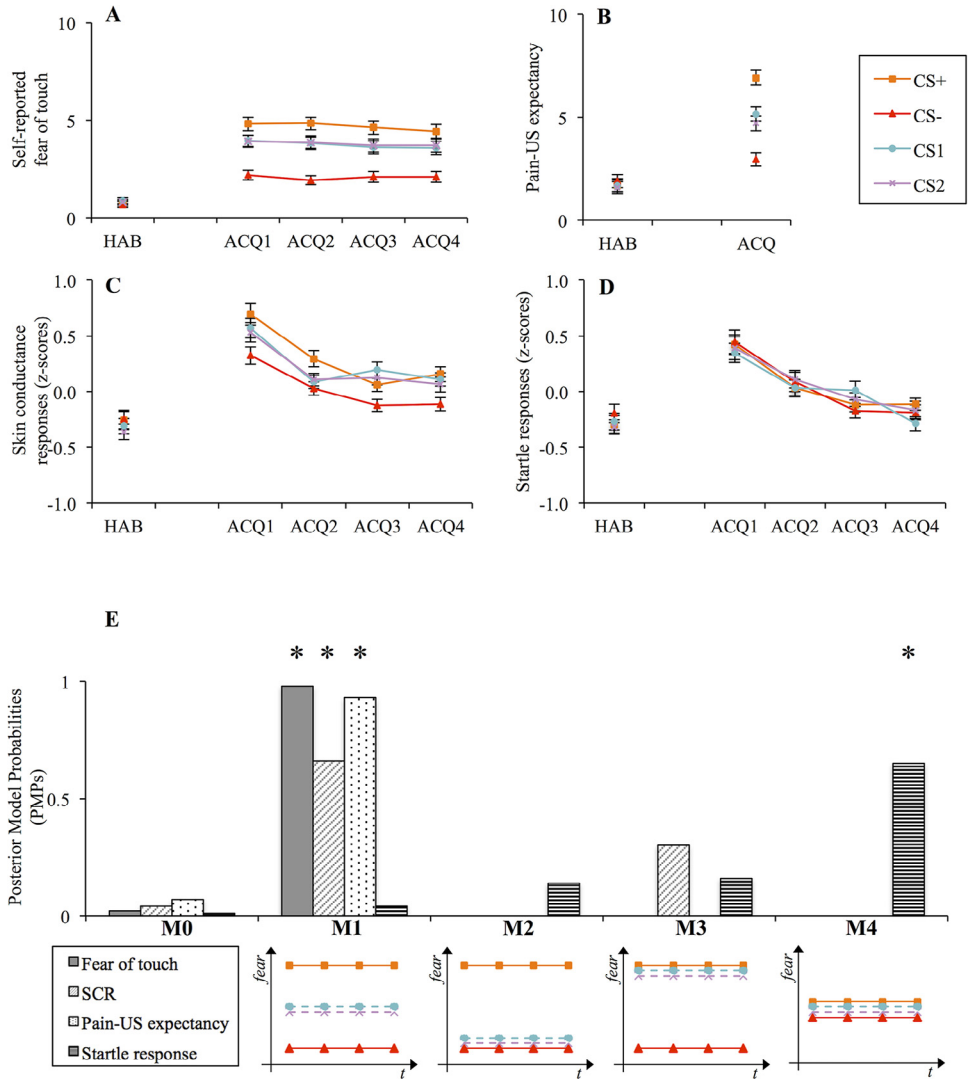

Figure 2. (**A–D**) Mean (± standard error) conditioned responses during acquisition (blocks 1–4) for all participants. (**E**) PMP from model selection. The probability of each model (depicted on the x-axis), for the conditioned responses shown in **A** through **D**, is summarized. The models compared the mean response to the CSs. *Indicates the most likely model for each measure. Abbreviations: HAB, pre-conditioning; ACQ, acquisition.

responses to all CSs). The alternative hypotheses all posited that fear responses would be higher for the CS+ than the CS–, whereas responses to CS1 and CS2 would be equal. However, one hypothesis specified that fear responses to the CS1 and CS2 would be higher than the CS– but lower than the CS+; the second hypothesis that fear responses to the CS1 and CS2 would be equal to the CS–; and the third that fear responses to the CS1 and CS2 responses would be equal to the CS+ (models depicted in Fig 2). Contextual fear responses (ITI startle responses) were hypothesized to be larger in the predictable than the unpredictable context, whereas an alternative hypothesis posited the reverse relationship. Two null hypotheses were included: no relationship between ITI startle responses in both contexts (M0) and equal ITI startle responses in both contexts.

Self-Reported Fear of Touch and Pain-US Expectancy

Participants were more afraid and more strongly expected the pain-US to occur during the CS+ than during the CS–. For the CSs in the unpredictable context (CS1 and CS2), no such differences occurred, but the fear and expectation of pain-US occurrence were higher for these CSs than the CS– (fear of touch: PMP = .98; pain-US expectancy: PMP = .93; Fig 2).

SCR

SCRs reflected the same pattern as the self-reported fear of touch and pain-US expectancy. Specifically, participants showed greater autonomic arousal to the CS+ than the CS1 and CS2 (which were equal) and all 3 CSs evoked greater responses than the CS– (PMP = .66; Fig 2).

Eyeblink Startle Response

The pattern found in the self-reports and SCR data, and that we had hypothesized, was not evident in the startle data. Participants showed equal startle responses to all CSs (PMP = .65; Fig 2). However, in line with our hypothesis about the effect of pain-US predictability on contextual fear, the average startle response during the

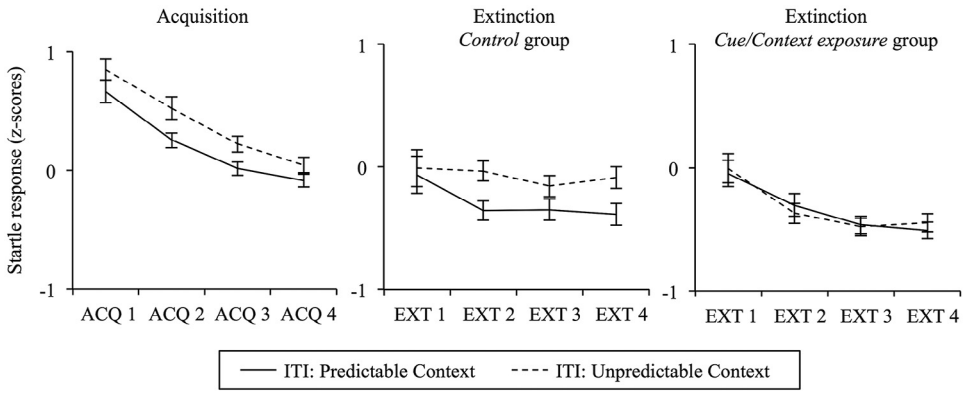

Figure 3. Average startle responses during the ITI in the predictable and unpredictable context. For acquisition (ACQ) the average of all participants is shown, for the extinction phase (EXT) the means per group are depicted.

ITI was elevated for the unpredictable compared with the predictable context (PMP = .67; Fig 3).

Extinction: Cue Exposure Group Versus Control Group

To assess the effect of the 'cue exposure' extinction protocol on fear responses, acquisition phase data were compared with data from the extinction phase, for the cue exposure group. The data from the predictable context of the control group served as a comparison to assess possible habituation or sensitization effects. Three hypotheses were tested for the difference in fear responses (self-reports and SCRs) between the CS+ and CS–: first, this difference could decrease from acquisition to extinction (hypothesis for the cue exposure group); second, this difference could increase; third, the differences remain unchanged (hypothesis for the control group). A null hypothesis (M0), that there is no relation between the CSs was also tested. The hypothesis for contextual fear was that ITI startle responses would be equal for the cue exposure group and the predictable context of the control group, 2 alternative hypotheses were that responses for the cue exposure group would be either greater or less than for the control group, whereas the null hypothesis was that there would be no relationship (M0).

Self-Reported Fear of Touch and Pain-US Expectancy

As expected, for the participants in the control group, fear of touch and pain-US expectancy ratings remained unchanged from acquisition (fear of touch: PMP = .46; pain-US expectancy: PMP = .43; Fig 4). For the cue exposure group, the CS+ no longer elicited more fear than the CS–, indicating that there was extinction of the previously established differential fear of touch (PMP = .66; Fig 4). In addition, the difference in pain-US expectation between CS+ and CS– also decreased (PMP = .62; Fig 4).

SCR

Successful extinction was also evident in the SCRs for the cue exposure group: the difference in SCRs to the CS+ and CS– was significantly reduced (PMP = .60), but also in the control group (PMP = .58; Fig 4). A post hoc test confirmed that this decrease was greater in the cue exposure group than in the control group (PMP = .54), suggesting that the decrease in the latter was due to habituation.

Eyeblink Startle Response

Because there were no differential responses to the CSs during the acquisition phase, the startle responses related to CS– were not compared in the extinction phase. To assess the effect of cue exposure on contextual fear, we compared the average startle response during the ITI in the predictable context for the cue exposure and the control group (Fig 3). As expected, there was no difference between the 2 groups (PMP = .48).

Extinction: Context Exposure Group Versus Control Group

To assess the effect of the 'context exposure' extinction protocol on fear responses, data from the acquisition phase were compared with data from the extinction phase, for the context exposure group. The data from the unpredictable context of the control group served as a comparison to assess possible habituation or sensitization effects. Three hypotheses were tested for the fear responses (self-reports and SCRs) to the CS1 and CS2: responses would remain unchanged from acquisition to extinction (hypothesis for the control group), responses would increase, or responses would decrease (hypothesis for the context exposure group). A null hypothesis (M0), that there was no relationship between CSs was also included. The hypothesis for contextual fear was that ITI startle responses would be smaller for the context exposure group than the unpredictable context of the control group, 2 alternative hypotheses were that responses for the cue exposure group would be either greater than or equal to those for the control group, whereas the null hypothesis was that there would be no relationship (M0).

Self-Report Fear of Touch and Pain-US Expectancy

Whereas the control group reported the same levels of fear of touch, and an equal expectation of pain-US occurrence (fear of touch: PMP = .75; pain-US

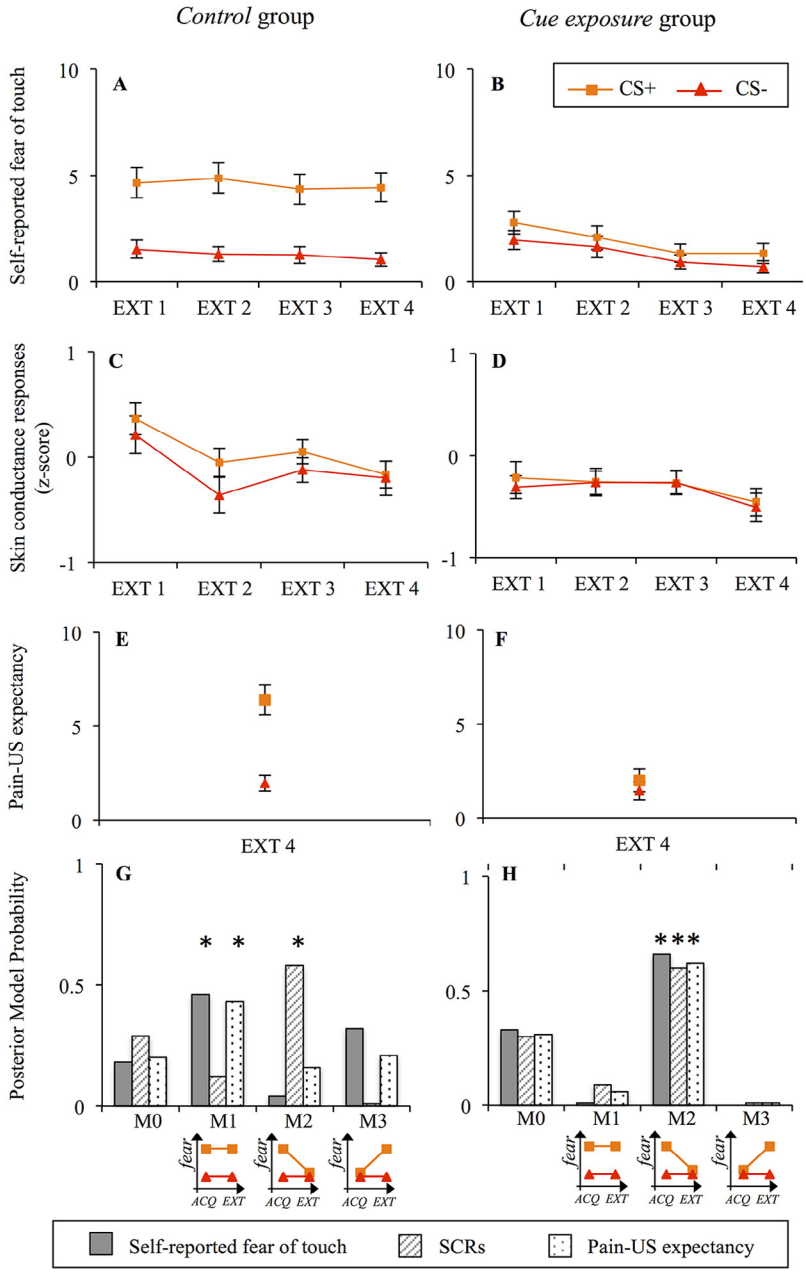

Figure 4. (**A–F**) Mean (± standard error) conditioned responses during the predictable context of the EXT (blocks 1–4), per group. (**G–H**) PMP from model selection. The probability of each model (depicted on the x-axis), for the conditioned responses shown in **A** through **F**, is summarized. The models compared the mean response during the ACQ phase with the mean response during the EXT. *The most likely model for each measure.

expectancy: PMP = .56; Fig 5) from acquisition to extinction, the context exposure group reported less fear during the extinction phase and also expected the pain-US to occur less (fear of touch: PMP = .80; pain-US expectancy: PMP = .80; Fig 5).

SCR

Autonomic arousal to the CS1 and CS2 remained unchanged for the control group during extinction (PMP = .51; Fig 5); whereas for the context exposure group the exposure protocol reduced the SCRs to the CS1 and CS2 (PMP = .80; Fig 5).

Eyeblink Startle Response

A comparison of the startle response during the unpredictable ITI for the context exposure and control groups showed, as expected, a decrease in contextual fear as a result of extinction (PMP = .85; Fig 3).

Discussion

The aim of this clinical analog study with healthy participants, was to investigate the acquisition and extinction of fear of touch as frequently observed among some chronic pain patients. Using a novel tactile fear conditioning paradigm, the effect of predictable pain occurrence (analogous

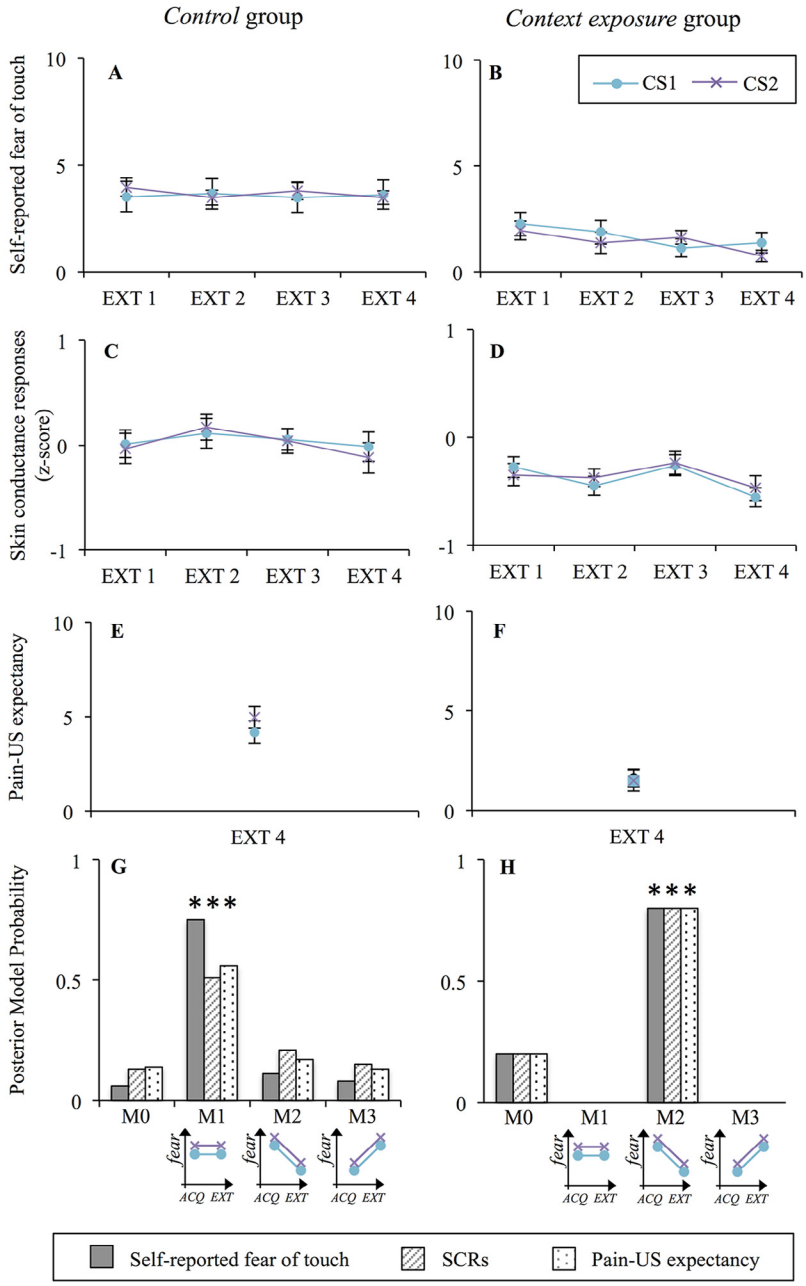

Figure 5. (**A–F**) Mean (± standard error) conditioned responses during the unpredictable context of the extinction phase (EXT; blocks 1–4), per group. (**G–H**) PMP from model selection. The probability of each model (depicted on the x-axis), for the conditioned responses shown in **A** through **F**, is summarized. The models compared the mean response during the acquisition phase (ACQ) with the mean response during the EXT. *The most likely model for each measure.

to the experience of allodynia) on fear of painful touch, as well as of unpredictable pain occurrence (analogous to the experience of spontaneous pain fluctuations) on contextual pain-related fear, were tested. The results indicate that the pairing of touch, in the form of vibrotactile stimulation, and pain can successfully trigger a conditioned fear response, evident from self-reported fear and increased autonomic arousal (SCRs), however, no potentiation of the startle response to any of the CSs was present. We found that when pain occurred unpredictably, as patients with spontaneous pain fluctuations may experience, there was potentiation of startle responses to the context (ie, during the ITI). Interestingly, this fear carried over to the CSs that occurred in the unpredictable context, because participants reported being more afraid of these CSs (CS1 and CS2) than the safe (CS–) stimulus in the predictable context. However, these CSs were functionally equivalent, because they were never paired with pain. It is possible that this increased fear was due to spillover effects, or that contextual fear impairs safety learning (for a similar discussion in fear of movement-related pain see the reports from Meulders and colleagues[29,30]). These results suggest that touch that simply occurs in the same context as unpredictable pain fluctuations may become feared and thus motivate maladaptive protective behaviors, such as avoidance.

After successful acquisition, a cue exposure and a context exposure protocol were used to reduce the fear of touch and contextual pain-related fear, respectively. In the cue exposure group, by exposing participants to the feared vibrotactile CS+, but without the pain reinforcement, self-reported fear and autonomic arousal decreased. In the context exposure group, where participants were exposed to the feared context without pain occurrence, there was a reduction in contextual fear, apparent from decreased startle responses compared with the control group. Furthermore, participants reported less fear and showed less autonomic arousal (decreased SCRs) in response to the CSs that occurred in this context, showing that context exposure was also successful in reducing the fear that was evoked by touch in this unpredictable context.

A clinical analog of these experimental extinction protocols is graded exposure in vivo (GEXP),[14] an intervention that aims to reduce functional disability in chronic low back pain and complex regional pain syndrome (CRPS) patients by challenging patients' beliefs about outcomes from feared activities, using behavioral experiments. It is grounded in the theory that pain-related fear can be acquired by associative learning, and the formation of new associations that inhibit fear responses reduce avoidance of feared activities and in turn also decrease disability.[13,40,41] Understanding how these pain-related fears develop is an important step to improving this therapy, because touch may become feared not because of its associated outcome, but because of the context in which it occurs. Therefore, targeting not just cued fears, but also contextual fears could be beneficial to optimally reduce fear and avoidance and thus its associated disability. Furthermore, although GEXP is effective, there are still some patients for whom the therapy is either not effective, or the effects do not persist. By testing variations to extinction protocols[31,34,35] within this paradigm, such as the effect of pain persistence and unconditioned stimulus-revaluation on fear of touch, it may be possible to identify strategies that will strengthen treatment outcomes or prevent relapse of pain-related disability.

Although our results generally corroborate previous findings using a fear of movement-related pain paradigm[29,30] the inhibited startle responses during tactile stimulation were unexpected. This inhibitory effect emerged before conditioning. It is possible that the onset of the tactile stimulation inhibited the startle response to the auditory probe, however, the startle probe occurred at least 1 second after the onset of the tactile stimulus, well beyond the usual range for prepulse inhibition (50–500 ms).[6,8] In only one of the previous studies that used tactile CSs[15,21,22,25,26] the eyeblink startle response was recorded.[22] In this study they found differential startle responses when the tactile CS+ and CS− were applied to the forearm. Unfortunately, no ITI startle response data were collected, and so it is not possible to compare whether a similar inhibitory effect was also present. Differences between this study and ours include the type of CS: we used 4 seconds of 30 Hz (flutter range) stimulation, whereas Harvie et al[22] used 5 seconds of 220 Hz (vibration range) stimulation. These different ranges use different mechanoreceptors, different afferents, and project to different cortical areas.[17,20,37] In addition, the startle response was elicited between 2 and 2.75 seconds after CS onset in the study of Harvie et al,[22] whereas in our study the earliest probe was at 1 second after CS onset, and the latest at 3 seconds after CS onset. If a delayed prepulse inhibition effect was responsible for the inhibition then the responses in the present study would be more greatly affected than those in Harvie et al.[22] Further research is clearly needed to investigate whether these physiological components and/or methodological choices can explain the difference in results.

Tactile fear conditioning has a huge potential to increase our understanding of fear of touch. The current paradigm was designed to specifically model symptoms experienced by many chronic pain patients, and therefore can be used to address research questions with clinical relevance. For example, it could be used to better understand the consequences of experiencing allodynia and spontaneous pain fluctuations in CRPS patients. This is a chronic pain condition characterized by diffuse pain in the upper or lower limbs.[10] In addition to pain-related symptoms such as allodynia and spontaneous pain,[7,12] the affected limb may also show edema, motor impairments, vasomotor (color and temperature), sudomotor (sweating), and trophic (nail and hair growth) changes.[19] A number of criteria have been proposed to assess the validity of an experimental model for a (subset of symptoms of a) clinical condition, including face validity, predictive validity, construct validity, and diagnostic validity.[39] Face validity requires that the experimental model and clinical condition share phenomenological similarity, in this situation the location of stimulation (fingers and wrists) in the paradigm are similar to those affected in CRPS. In addition, both types of pain experience are modeled (stimulus-evoked and stimulus-independent). Expansion of the paradigm could further satisfy this criterion by including manipulations to mimic other symptoms also experienced by CRPS patients, or by using a tonic pain stimulus to induce a longer-lasting sensation. However, the aim of creating an experimental model is also to simplify and control various features that are too complex to study in their clinical reality and, as noted by Vervliet and Raes,[39] "face validity is neither a necessary nor sufficient criterion of external validity." The second criterion, predictive validity, refers to the degree to which behavior in the experimental model can be used to predict behavior in the clinical condition. From this first study, the predictive validity appears strong; the reduction in fear responses as a result of the exposure manipulation is in line with the reduction in fear reported by CRPS patients after GEXP therapy.[14] The third criterion, construct validity, requires that the experimental model is simulating the same etiological process as is occurring in the clinical condition. In this experimental model, the mechanism underlying the development and reduction of the conditioned fear response is assumed to be associative learning, research with CRPS patients is currently lacking as to whether or not fear of touch is also acquired in the same way in this group. The final

criterion is diagnostic validity: can patients be distinguished from healthy controls by deviant behavior on the same task? To address this criterion the paradigm would need to be tested with CRPS patients. However, similar research with fibromyalgia patients,[28] and unilateral hand pain patients,[27] does suggest that deficiencies in safety learning could be expected when comparing patients and controls. On the basis of these criteria, the paradigm used in this study is a first step toward an experimental model having the potential of strong clinical relevance that can be translated to chronic pain syndromes.

Conclusions

Our paradigm aimed to model the fear evoked by 2 types of pain symptoms in chronic pain conditions: fear of touch due to allodynia and contextual pain-related fear due to unpredictable pain fluctuations. Our paradigm was successful in eliciting both states and shows how the autonomic nervous system responds to not only touch that is associated with pain, but also touch that simply occurs in the same context as pain. In line with our expectations, cue exposure proved to be a successful method in reducing fear of touch and context exposure reduced contextual pain-related fear, a first step toward investigating processes that may help us to better understand how these fears are acquired, and can be reduced. This line of research is necessary to increase the efficacy of GEXP and other exposure-based treatments for chronic pain patients suffering from fear of touch.

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Supplementary Data

Supplementary data related to this article can be found online at http://dx.doi.org/10.1016/j.jpain.2017.08.002.

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